Explosive Detector



Category: **Safety & PPE** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Explosive Vapour Detector (EVD)

What it is used for

An explosive vapour detector checks the surrounding atmosphere for vapours or gases associated with specified explosive or hazardous substances, depending on its sensor technology. Correct calibration checks, approved sampling methods, and careful interpretation help identify potential hazards without treating a negative reading as proof that an area is safe.

Main parts

- Detector unit and display
- Sensor or sampling inlet
- Sampling probe or tubing where fitted
- Battery or approved power supply
- Alarm indicators and audible alert
- Calibration accessories and manufacturer-approved test materials

Before you start

1. Use the detector only if trained and authorised for the specific instrument and site safety procedure.
2. Confirm the detector's intended target substances, sensor technology, detection limits, and operating environment from the manual.
3. Inspect the unit, sensor inlet, sampling tubing, battery, alarm indicators, and housing for damage or contamination.
4. Verify calibration status, sensor service life, and the required self-test or bump-test procedure.
5. Perform the prescribed functional check using only manufacturer-approved test materials and methods.
6. Review site emergency procedures, required PPE, access restrictions, and communication arrangements before entering the test area.

How to use

1. Switch on the detector in the designated safe area and allow it to complete its startup and self-test sequence.
2. Confirm that the display, alarms, battery status, and sensor readiness indicators show normal operation.
3. Perform the required bump test or calibration check according to the manual and record the result.

4. Select the correct detection mode or target setting when the instrument supports multiple modes.
5. Approach the inspection area according to the site's approved hazardous-material response procedure and maintain a safe position.
6. Sample the atmosphere at the locations and heights specified by the instrument manual and site procedure, allowing the required response time.
7. Move the sampling inlet slowly and avoid blocking it against surfaces or exposing it to liquids, dust, or incompatible materials.
8. Monitor readings, trend changes, and visual or audible alarms continuously during the survey.
9. If an alarm activates or a suspicious reading appears, stop the survey, avoid further exposure, and follow the site's evacuation and incident-response procedure.
10. Do not touch, collect, open, or move any suspected explosive material; leave investigation and handling to the designated specialist team.
11. Record the location, time, detector identification, readings, alarm events, and environmental conditions as required.
12. After the survey, move to the approved safe area and follow the manufacturer's procedure to shut down and inspect the detector.
13. Clean the exterior and sampling accessories only using approved methods and store the unit securely.
14. Report abnormal readings, failed checks, suspected contamination, or sensor faults to the responsible safety officer.

⚠ Safety

- A detector reading is an aid to risk assessment, not a guarantee that an area is safe; follow the site hazard assessment and specialist instructions.
- Never approach or handle a suspected explosive device or suspicious package to obtain a reading.
- If an alarm or suspected explosive hazard is encountered, do not investigate further; withdraw to a safe location and notify the designated emergency or specialist team.
- Use only detectors approved for the target hazard and environment; a general-purpose vapour detector may not detect every explosive substance.
- Do not use unapproved solvents, test chemicals, or improvised calibration methods on the sensor.
- Observe hazardous-area restrictions and use only equipment approved for the area's ignition-risk classification.
- Do not enter a potentially hazardous area alone or without the required PPE, communication, and emergency arrangements.

Care & storage

- Clean the detector exterior and sampling inlet according to the manufacturer's instructions after use.
- Inspect tubing, filters, sensors, seals, and connectors for wear, blockage, or contamination.
- Charge or replace batteries as specified and confirm adequate power before deployment.

- Maintain calibration, bump-test, maintenance, and sensor-replacement records.
- Store the detector in its approved case under the specified temperature, humidity, and transport conditions.

Common mistakes to avoid

- Assuming a negative reading proves that no explosive hazard is present.
- Using a detector that is out of calibration, has a failed sensor, or has insufficient battery power.
- Sampling at inappropriate locations or failing to allow the sensor response time.
- Approaching, touching, or attempting to move a suspected explosive object instead of withdrawing and reporting it.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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